Second Supplemental Testimony of Gas Infrastructure and Operations Panel

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Second Supplemental Testimony

of

Gas Infrastructure and Operations Panel

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Case 19-G-0309
Case 19-G-0310

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Second Supplemental Testimony of Gas Infrastructure and Operations Panel

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I. Introduction

Q. Please identify the members of the Gas Infrastructure and Operations Panel.

A. The Panel consists of Ross W. Turrini, Timothy S. Graham, Caroline Hon, and Srividya Madhusudham.

Q. Is this the same Panel that previously submitted testimony on behalf of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, the "Companies") in these proceedings?

A. Yes. Previously-defined terms have the same meanings here.

II. Purpose of Testimony

Q. What is the purpose of the Panel's second supplemental testimony?

A. The purpose of the Panel's second supplemental testimony is to (i) update the Companies' capital and O&M expense forecasts to incorporate changes necessary to implement the recent settlement agreement between the Companies and DPS Staff in Case 19-G-0678 concerning the gas supply constraints in downstate New York and related moratorium on new customers connections, and (ii) update the Companies' cost of service to reflect the impact of contract negotiations with certain contractors.

Q. Does the Panel's second supplemental testimony include any exhibits?

A. Yes. The Panel's second supplemental testimony includes the following exhibits that were prepared or compiled under our direction and supervision:

- Exhibit ___ (GIOP-1SS): KEDNY and KEDLI's Revised Capital Investment Plans Reflecting New Customer Connections
- Exhibit ___ (GIOP-2SS): KEDNY and KEDLI's Revised O&M Plans Reflecting New Customer Connections
- Exhibit ___ (GIOP-3SS): KEDNY and KEDLI's Revised Proposed Incremental FTEs
- Exhibit ___ (GIOP-4SS): KEDLI's Revised Proactive Main Replacement (LPP) Unit Costs for FY2020-FY2024
- Exhibit ___ (GIOP-5SS): Worksheets Summarizing Changes to Capital Plan
- Exhibit ___ (GIOP-6SS): Worksheets Summarizing Changes to O&M Plan

III. <u>Updates to Incorporate the Settlement in Case 19-G-0678</u>

Q. Please summarize how the Companies presented their capital and O&M investment plans in these filings.

A. The Companies direct and corrections and updates testimony and exhibits were prepared under the assumption the NESE project would be in-service by the winter of 2020/2021, thereby allowing the Companies to continue connecting new customers. To provide the Commission with a full and complete evidentiary record in these proceedings, the Companies filed supplemental testimony on June 11, 2019, reflecting a cost of service under a no-NESE project scenario where the Companies would not be connecting new customers during the Rate Year and Data Years.

Staff, in its direct testimony, took the position that the Companies' capital and investment plans should be adjusted to assume the unavailability of the NESE project. Accordingly, in rebuttal, the Companies revised their capital and O&M investment plans to assume the NESE project would not be in service and that new customers would not be connected in the Rate Year or Data Years.

Q. Please describe the recent settlement in Case 19-G-0678 and how it impacts these rate filings.

A. On November 26, 2019, the Commission approved a settlement agreement between the Companies and DPS Staff in Case 19-G-0678 ("Settlement")

where the Companies agreed to lift the moratorium for the next two winters (through September 1, 2021). To enable new customer connections, the Companies have developed an operational plan and supply solution that provides for the expanded use of portable compressed natural gas ("CNG") injections to maintain system pressure during peak conditions, as well as various demand-side measures and the continued reliance on city gate peaking supplies. The Panel is supplementing its testimony and exhibits to reflect the impacts of the Settlement on various programs and cost components in the Companies' capital and O&M expense forecasts.

Specifically, the Companies are updating the capital forecasts for the Customer Connections programs to reflect the costs of new customer additions (installing main and services, meters, and AMR requirements) and the Gas System Reinforcements and Reliability projects necessary to maintain system pressures and serve additional load (*e.g.*, transmission upgrades). To a large extent, these costs were included in the Companies' initial filing on April 30, 2019 and updated in the Companies' July 3, 2019 corrections and updates filing.

The Companies are also updating their capital and O&M plans to reflect the costs of projects that are required to provide the additional on-system supply

necessary to meet peak demand, including new CNG and liquified natural gas ("LNG") facilities and upgrades. These investments are critical to enabling new customer connections and are expressly contemplated in the Settlement.

The updates are reflected in Exhibit __ (GIOP-1SS) and Exhibit __ (GIOP-2SS) and are described in detail below.

Q. Please describe the Companies' updates to the Customer Connections forecasts.

A. KEDNY and KEDLI's capital forecasts for the Install Main, Install Services, Meter Purchase, Install Meter/Regulator, and AMR line items have been updated to reflect the costs to connect new customers as agreed in the Settlement. These forecasts largely are consistent with the Customer Connections budgets presented in the Companies' corrections and updates filings.

Additionally, KEDLI has updated its Meter Purchases, Install Meter/Regulator, and AMR forecasts to remove costs associated with the Clean Choice Program because it is no longer proposing this gas conversion program. As updated in Exhibit __ (GIOP-1SS), the Meter Purchases,

Install Meter/Regulator and AMR forecasts now reflect meter and AMR requirements for new customer connections during the Rate Year and Data Years.

Q. Please describe the updates required to KEDNY's Gas System Reinforcement and Reliability programs to enable new customer connections.

A. The Companies' rebuttal filing reflected adjustments to KEDNY's Gas System Reliability Program due to the anticipated moratorium on new customer connections. The rebuttal filing also contemplated the addition of the Jamaica Inlet Project that was required to address low system pressures in the absence of the additional supply from NESE.

Now that the moratorium has been lifted, KEDNY's Gas System Reinforcement Program has been updated to remove the moratorium-related adjustments, thus restoring the forecast costs for this program to the amount reflected in the corrections and updates filing.

However, KEDNY's MRI Reliability project must be accelerated by one year to provide the hydraulic system support that is critical for enabling new customer connections over the next few winters. Exhibit __ (GIOP-1SS)

reflects the acceleration of the MRI project. Acceleration of the MRI project avoids the need for the Jamaica Inlet Project, which has been removed from KEDNY's capital forecast. Additionally, the Marine Park Regulator Station Program has been removed from the forecast because it was only needed to support the NESE project.

Additionally, over the longer term KEDNY requires one new system reliability project to support new customer connections. The Clove Lakes Transmission Loop Project has been added to the FY 2021 and FY 2022 capital plans to enable KEDNY to conduct a feasibility study and engineering analyses. Looping of the Clove Lakes Transmission Main with a new 30-inch transmission main across the borough of Staten Island will facilitate KEDNY's ability to take more gas through the TETCO Goethals Take Station and ultimately move this gas across KEDNY's system to supply constrained areas. This project will increase supply diversity and delivery capacity for long term growth to meet design day conditions as well as improve overall system reliability. This project is essential to maintaining pressure on the transmission system as part of KEDNY's longer-term plans to support new customer connections but is not expected to be in service until FY 2025, at the earliest. The forecast estimates for

conducting the feasibility and engineering studies are $0.5 million in FY 2021 and $1.5 million in FY 2022.

Q. Please describe the updates required to KEDLI's Customer Connections – Gas System Reinforcement Program and Reliability category programs to enable new customer connections in the absence of the NESE project.

A. KEDLI has also updated its Gas System Reinforcement Program forecast to remove the moratorium-related adjustments that were presented in the Companies' rebuttal filing. The forecast is restored to the version that was included in the corrections and updates filing to reflect reinforcements required to connect new customers.

Additionally, KEDLI's updates include adding back the Riverhead Transmission and Southeast Suffolk Phase 2 projects that were removed in the rebuttal testimony. These projects are required to support new customer connections in eastern Long Island. Moreover, to enable new customer connections, KEDLI has accelerated Phase 1 of the Southeast Suffolk project to ensure it will be in service in FY 2022. Accordingly, Exhibit __ (GIOP-1SS) reflects a shift of $5 million of the project forecast from FY 2022 to FY 2021.

Q. Are the Companies updating the costs for the Enhanced Contractor Inspection Program?

A. Yes. In the Companies' rebuttal filing, the Companies reduced the number of contractor inspectors required for this program because there would be a reduction in contractor crews due to the moratorium on new connections, and thus fewer inspectors were needed to meet the Companies' goal of a one-to-one inspector to crew ratio. Now that the moratorium is no longer in place, the contractor crew reduction will not occur. Thus, the Companies have adjusted the capital and O&M forecasts to the same levels that were submitted in the corrections and updates filing.

Q. Please explain the investments in CNG facilities that are required to support service to new customers.

A. The Companies' interim plans to ensure the peak day supply needed to serve new customers rely on the expanded deployment of CNG. There are two existing CNG injection sites in KEDLI's service territory at Glenwood and Riverhead (available January 2020), which will be utilized to serve peak demand for the winter 2019/2020.

For the winter 2020/2021, to meet increased gas demand from new and existing customers, the Glenwood CNG facility will be expanded to double

its capacity. Two additional CNG injection facilities (a third and a fourth CNG facility) will be required in KEDLI's service territory at locations to be determined. At this point, five potential properties that are owned by National Grid in KEDLI's service territory are being considered. The capital costs for each new facility are estimated at approximately $1 million in FY 2020 and $10 million in FY 2021. These costs are reflected in Exhibit __ (GIOP-1SS). The new facilities will also require incremental O&M expense that is included in Exhibit __ (GIOP-2SS).

Finally, the Companies require a CNG Filling Station to be located on Staten Island where trucks will be filled with CNG for transportation to both KEDLI's and KEDNY's CNG injection sites. The total cost for the project is approximately $30 million, and the project is expected to be in service in March of 2022. The cost is being reflected in KEDNY's capital investment forecast in Exhibit ___ (GIOP-1SS) because the asset is being built in KEDNY's service territory; however, KEDLI will pay 50 percent of the cost to KEDNY as a rental because the asset equally benefits KEDNY and KEDLI. KEDLI's rental expense related to this project is reflected in Exhibit ___ (GIOP-6SS) and KEDNY's rental income related to this project is reflected in Exhibit __ (RRP-3SS) and is included in the revenue requirement accordingly.

Q. Are updates required to the Companies' LNG investments to support new customer connections?

A. Yes. The Companies' plans for interim solutions to the capacity constraints involve additional reliance on the Companies' LNG capabilities, including updates to the Companies' existing LNG Tank Upgrade programs and to KEDNY's LNG Truck Load/Unload Station Program. Additionally, new investments are required for KEDNY's Greenpoint LNG site to add: (i) a new portable LNG facility and (ii) two additional vaporizers.

Q. Please describe the updates to the LNG Tank Upgrade Programs.

A. The Companies' LNG Tank Upgrade Programs as submitted in the April 30 rate case filing were initially designed as full replacements of the tanks. This project would require the tanks to be taken out of service for an extended period, including the winter heating season. As has been stated in the Companies' prior submissions, it is not possible to take the tanks out of service during the winter absent the additional supply capacity. However, considering the age of the tanks and the importance of reliable operation of the Companies' LNG facilities, particularly now that the Companies will be adding new customers over the next several winters, the Companies have designed a more streamlined and targeted approach to performing inspections and repairs on the tanks. KEDNY's Greenpoint LNG Tank 2

will be prioritized over Tank 1 because of known condition issues related to the foundation heating system. Tank 2 will be taken out of service from April of 2024 to October of 2024 to perform inspection and any necessary repairs so that the tank will be in service for the winter heating season of November 2024 to March 2025. Tank 1 will be addressed beginning in FY 2026. This modified work scope will ensure that the tanks are in service for the winter heating seasons.

KEDLI's Holtsville LNG Tank will be taken out of service during the Spring of 2023 and will be back in service by November of 2023.

Updates to the forecasts for the Greenpoint Tank 2 Upgrade Program, removal costs of the Greenpoint Tank 1 Upgrade Program, and updates to the Holtsville Tank Upgrade Program are reflected in Exhibit __ (GIOP-1SS).

Q. Please describe the update to KEDNY's LNG Truck Load/Unload Program.

A. KEDNY's Truck Load/Unload Program is a special project that is described in the April 30 filing on page 101 of Exhibit __ (GIOP-5). The program provides capability to transfer portable LNG in the event of an unplanned

outage at the Greenpoint liquefaction unit. The project was planned to begin in FY 2020; however, the project has been delayed because KEDNY and the City of New York have not yet executed a Memorandum of Understanding ("MOU") that would allow KEDNY to transport LNG via trucks through New York City. In light of the increased role LNG capabilities will play in enabling KEDNY to deliver reliable service to new and existing customers, the Company anticipates that the MOU will be executed in the near future. Thus, KEDNY's updated capital plan reflects delaying the investment in this project from FY 2020 to FY 2021 and compressing the remaining work plan to ensure that the station is in service by FY 2023 to remain on schedule to ensure reliability. Compression of the work plan requires an additional $2.44 million in project costs to cover expediting fees for faster equipment manufacturing.

Q. Please explain the new investments required for KEDNY's Greenpoint LNG site.

A. KEDNY's plan to ensure the continued ability to support customer load includes the addition of portable LNG capabilities at the Greenpoint site that will allow LNG delivered via truck to on-system injection points. The project is anticipated to be in service in November of 2022 and requires a capital investment of $0.5 million in FY 2020 and $4.5 million in FY 2021

for permitting, engineering and site work, and investments of $7.5 million in FY 2022 and $15 million in FY 2024 for construction.

Additionally, KEDNY's original rate case filing included a project to replace vaporizers 3 and 4 that will be complete in the Spring of 2020 and that will enable delivery of increased peak hour and peak day supply from the Greenpoint plant for the winter of 2020/2021 (the new vaporizers replacing 3 and 4 will be named 11 and 12 to avoid confusion on system schematics and drawings). To support increased customer demand beyond the 2020/2021 winter, KEDNY needs to add two additional vaporizers to the Greenpoint site, which will be Vaporizers 13 and 14. The LNG facility has enough pump capacity for Vaporizers 13 and 14. Each vaporizer is capable of outputting 2,500 Dth/hr of natural gas. Over a twenty-four hour period, this equates to 60,000 Dth per day (they are not run simultaneously). By comparison, a similar CNG project proposed in Greenpoint would decompress CNG at a rate of 2,200 Dth/hr up to 6,400 Dth per day. For a CNG injection facility to match the daily output of the vaporizers, it would require 143 CNG trailers. Vaporizers 13 and 14 are expected to be in service in FY 2023, and the costs of $1 million in FY 2020, $13 million in FY 2021, $21 million in FY 22, and $20 million in FY 23 are included in Exhibit __ (GIOP-1SS).

IV. Impact of Recent Contractor Agreements

Q. Please explain why the Companies are updating contractor costs.

A. The Companies have completed contract renegotiations with certain contractors that will provide necessary services in the Rate Year and Data Years. These negotiations have resulted in known and measurable decrease to the unit costs of KEDLI's Proactive Main Replacement (LPP) Program and an increase to the damage prevention mark out costs for both Companies, and, therefore, require updates to the capital and O&M expense forecasts.

Q. Please describe the update to the forecast for KEDLI's Proactive Main Replacement (LPP) Program.

A. KEDLI renegotiated its agreement with a contractor that provides services for the Proactive Main Replacement (LPP) Program. A copy of the renegotiated agreement was provided in the Company's response to IR No. DPS-1043.

Q. How does the agreement impact KEDLI's costs for this program?

A. The agreement results in reduction of the program's unit costs from $306 per foot to $304 per foot, as shown in Exhibit ___ (GIOP-4SS). Revising

the cost per foot reduces KEDLI's capital expense forecast by $1.929 million in the Rate Year, as shown in Exhibit ___ (GIOP-1SS).

Q. Please explain damage prevention mark outs and how the costs of these mark outs are included in the Companies' rate filings.

A. The Companies' Damage Prevention group performs mark outs of the Companies' underground utilities to allow for safe excavation and prevent damage to existing facilities, as required by the Commission's regulations. Damage prevention mark outs are performed when required to support one of the Companies' projects or when customers, third parties, and municipalities contact New York State's Dig Safe One Call Center. KEDNY uses internal employees to locate facilities on Company-related jobs and a contractor for all third-party Dig Safe mark out requests. Conversely, KEDLI sends all Dig Safe mark out requests, whether internal or external, to a contractor to mark out facilities.

The costs associated with damage prevention mark outs are included in KEDNY's O&M expense and KEDLI's O&M and capital expense. To prepare the rate case forecasts, the Companies increased these expense components from the Historic Test Year by standard annual inflation to arrive at the Rate Year forecast.

Q. Please explain recent developments that will affect the contractor cost component of damage prevention mark out costs.

A. On August 19, 2019, the Companies solicited contractor bids for their damage prevention mark out work. Bids were received from contractors on September 16, 2019. Thereafter, bidders were evaluated both commercially and on their ability to execute technical requirements. Extensive negotiations were also conducted with bidders between September 16 and November 8.

Q. Have the Companies selected contractors based on the bids received?

A. Yes. The Companies' are in the process of executing agreements with two contractors to provide damage prevention mark out services from February 2020 to December 2022. The agreements are expected to be executed on or about January 1, 2020. The Companies' are therefore updating the capital and O&M forecasts to include these increased costs.

Q. What effect will the updated agreements have on the Companies' contractor costs?

A. The contracts will raise the labor rates for these contractors in excess of the standard annual inflation rate used to derive the Companies' Rate Year forecasts.

Q. What are the drivers of the higher contractor bids?

A. The increase is attributable to three factors: (i) the prior contractor labor rates were based on long-term, dated agreements; (ii) recent unionization of the contractor labor force; and (iii) an increased scope of work to incorporate locator assurance technology. The Companies have not bid this work in over a decade. While the prior agreements provided for minimal price increases, the rates paid to those contractors were generally lower than market prices at the time. Contractors were no longer willing to accept below-market rates during the recent negotiations. Additionally, the contractors providing services under the selected agreements unionized during the late Spring and early Summer of 2019. While the Companies were generally aware that these labor forces were unionizing, the increase in labor rates experienced was significantly higher than anticipated or reflected in the Companies' initial filing. In addition, the new agreements require that the contractors use updated technology (locator assurance technology that enables real-time tracking of technician performance against industry standards to improve overall effectiveness of locating field work) when marking out facilities. This requirement expands the scope of work required in prior agreements and will require additional training and inventory for these contractors.

Q. What are the overall impacts of the changes to the contracts for damage prevention mark-out services to the Companies' capital and O&M forecasts?

A. Tables 1 and 2 below present summaries of the impacts of these agreements on the Companies' capital and O&M forecasts. These impacts are also reflected in the Damage Prevention line item in the updated Exhibit ___ (GIOP-1SS) and Exhibit ___ (GIOP-2SS).

Table 1: KEDNY Damage Prevention Mark Out O&M Updates FY2020 to Data Year 3

($000)	FY2021	FY2022	FY2023	FY2024
Original Forecast	7,588	7,979	8,445	9,005
Revised Forecast	14,417	14,755	15,102	15,460
Variance	6,829	6,776	6,657	6,455

Table 2: KEDLI Damage Prevention Mark Out Capital and O&M Updates FY2020 to Data Year 3

($000)	FY2021		FY2022		FY2023		FY2024	
	Capital	O&M	Capital	O&M	Capital	O&M	Capital	O&M
Original Forecast	3,315	1,637	3,384	1,766	3,455	1,914	3,528	2,083

Revised Forecast	8,339	3,574	8,339	3,574	8,339	3,574	8,339	3,574
Variance	5,024	1,937	4,955	1,808	4,884	1,660	4,811	1,491

Q. Did the Companies consider alternatives to use of contractors given the increase in costs for damage prevention mark out work?

A. Yes. The Companies conducted an analysis on whether it would be less expensive if this work was performed in-house. The results of that analysis determined that outsourcing the work was the least cost option. Procurement also performed an analysis that compared all bids received based on various criteria and determined that the contractors selected were the best overall option.

V. Conclusion

Q. Does this conclude the Panel's second supplemental testimony?

A. Yes.

INDEX OF EXHIBITS

Exhibit __ (GIOP-1SS): KEDNY and KEDLI's Revised Capital Investment Plans Reflecting New Customer Connections

Exhibit __ (GIOP-2SS): KEDNY and KEDLI's Revised O&M Plans Reflecting New Customer Connections

Exhibit __ (GIOP-3SS): KEDNY and KEDLI's Revised Proposed Incremental FTEs

Exhibit __ (GIOP-4SS): KEDLI's Revised Proactive Main Replacement (LPP) Unit Costs for FY2020-FY2024

Exhibit __ (GIOP-5SS): Worksheets Summarizing Changes to Capital Plan

Exhibit __ (GIOP-6SS): Worksheets Summarizing Changes to O&M Plan

Exhibit ___ (GIOP-1SS)

KEDNY and KEDLI's Revised Capital Investment Plans
Reflecting New Customer Connections

The Brooklyn Union Gas Company d/b/a National Grid NY
Direct Capital Expenditures (CAPEX and COR)

	FY20	FY21	FY22	FY23	FY24
Customer Connections					
Customer Connections - Install Main	21,146,720	21,729,722	22,538,940	22,989,719	23,449,513
Customer Connections - Install Services	24,785,380	25,488,092	27,925,611	28,484,123	29,053,806
Customer Connections - Customer Contributions	(2,307,000)	(2,352,000)	(2,403,000)	(2,456,000)	(2,503,000)
Customer Connections - Meter Purchases	1,811,750	1,847,990	1,884,950	1,922,640	1,961,100
Customer Connections - Install Meter/Regulator	1,232,673	1,257,700	1,336,904	1,363,642	1,390,915
Customer Connections - Automatic Meter Reading (AMR)	1,042,000	1,062,090	1,083,330	1,105,000	1,127,100
Gas System Reinforcement	45,382,000	13,641,000	61,716,000	84,342,000	64,031,000
LTNY11751 - Kew Gardens Gate - PM	4,806,891	17,937,000	-	-	-
LTNY12025 - Belmont Gate Station - PM	-	-	180,000	720,000	25,514,000
Total Customer Connections	97,900,413	80,611,594	114,262,735	138,471,124	144,024,433
Mandated					
CSC/Public Works - Non Reimbursable	122,011,065	125,897,715	132,501,150	134,914,950	125,596,350
CSC/Public Works - Reimbursable	149,124,635	153,874,985	161,945,850	164,896,050	153,506,650
Flatlands - SE853 Phase 2 - Trans Offset Louisiana Ave & Georgia Ave	69,416,000	-	-	-	-
SE856 Phase 2 Trans. Offset Sheffield & New Jersey Ave. - Trans Work	1,609,000	26,590,000	26,980,000	27,590,000	4,000,000
SE856 Phase 2 Trans. Offset Sheffield & New Jersey Ave. - Dist Work	-	14,400,000	14,400,000	-	-
LaGuardia Redevelopment	654,382	164,382	-	-	-
CSC/Public Works - Reimbursements	(42,285,161)	(33,399,619)	(34,102,677)	(35,788,913)	(36,452,974)
Main Replacements - (Proactive) - Leak Prone Pipe	196,552,000	250,061,000	304,804,000	347,927,000	407,571,000
CISBOT	5,236,499	5,336,499	5,400,000	5,500,000	5,600,000
Large Diameter Main Rehabilitation	13,620,628	14,088,000	14,376,000	14,671,000	14,975,000
Cross Bore Remediation	396,839	150,000	153,000	156,060	159,181
Latent Damage Inspections	408,000	416,000	424,000	432,000	440,640
Main Replacements - (Reactive) - Maintenance	5,336,797	6,941,127	7,184,715	7,348,454	7,497,927
Service Replacements - Proactive	1,961,847	2,053,847	2,239,000	2,275,000	2,320,500
Service Replacement (Reactive) - Leaks	5,049,905	5,148,762	5,350,989	5,469,719	5,574,019
Service Replacement (Reactive) - Non-Leaks - Other	5,116,495	5,216,717	5,424,897	5,545,267	5,651,008
Atmospheric Corrosion Inside Inspections	100,000	650,000	104,000	106,000	108,000
Restrictions for Elevated Gas Infrastructure	336,000	373,000	381,000	388,000	396,000
Buried Vent Lines	108,000	111,000	113,000	115,000	117,000
Plastic Fusion QA/QC Re-Digs	3,260,000	3,250,200	3,391,704	3,459,538	3,528,728
Plastic Fusion - In Process Inspections	301,500	307,530	313,680	319,954	326,353
Low Pressure Main Valve Installation	-	2,460,000	2,723,000	2,956,000	3,196,000
High Density Polyethylene Services	-	2,458,800	2,520,270	2,583,277	2,647,859
Contractor Safety Inspections	-	5,370,628	16,363,614	27,786,267	28,182,040
Operator Qualification Program		909,361	519,653	529,827	541,029
Local Law 30	37,200,000	11,400,000	-	-	-
Inactive Accounts	268,924	274,924	287,000	293,000	299,000
Corrosion	927,028	1,004,571	994,571	983,769	1,066,059
Pipeline Integrity - IMP	-	500,000	1,501,350	-	-
Pipeline Integrity - IMP - Jamaica Bay Line IL	-	2,000,000	10,000,000	10,000,000	10,000,000
Pipeline Integrity - IMP - Southern Line Robotic IL	3,002,700	3,000,000	10,000,000	10,000,000	18,000,000
Pipeline Integrity - IVP	2,238,083	3,050,000	4,700,000	4,000,000	5,000,000
Pipeline Integrity - IVP Reactive Main Replacement	-	500,000	510,000	520,000	530,604
Valve Installations/Replacements	142,000	142,000	146,000	146,000	149,000
Meter Changes	4,328,998	4,437,998	4,593,000	4,708,000	4,825,000
Purchase Meters (Replacements)	3,662,544	3,736,114	3,826,940	3,903,480	3,981,550
Transmission Station Integrity	180,000	3,000,000	17,000,000	17,340,000	17,687,000
Complex Capital Delivery Initiative - Savings	-	(577,500)	(2,663,850)	(1,367,000)	(1,784,350)
Total Mandated	590,264,707	625,298,042	724,405,857	769,707,698	795,236,172
Reliability					
I&R - Reactive	514,743	524,484	527,241	538,940	549,217
I&R - Training and Test Lab	-	800,000	1,000,000	-	-
Gas System Control	114,852	117,182	121,120	123,540	126,010
Gas System Control - Telemetry Upgrade 3G to 4G	198,977	-	-	-	-
Heater Installation Program	-	500,000	2,500,000	750,000	750,000
Pressure Regulating Facilities	1,400,000	7,050,000	10,100,000	7,175,000	10,450,000
System Automation	1,394,307	1,394,307	1,700,000	1,734,000	1,734,000
Shafer Narrows	200,000	-	-	-	-
Bowery Bay Station Upgrade	100,000	500,000	3,500,000	300,000	-
McGuiness Mini Gate	-	-	250,000	3,500,000	250,000
Kings Plaza Mini Gate	-	-	250,000	3,500,000	500,000
Bush Terminal (IF-09)	-	-	-	250,000	3,600,000
Tetco Relief Valve Replacement	6,400,000	-	-	-	-
Citizens Gate - Bulkhead	7,060,000	3,100,000	-	-	-
Sheepshead Bay Mini Gate	-	-	-	200,000	3,600,000
GOV 110	149	-	150,000	3,200,000	-
Hyman station	-	300,000	3,500,000	250,000	-

Varick Reg Station Retirement	1,624,000	-	-	-	-
North Brooklyn Mini Gate	2,100,000	3,800,000	300,000	-	-
Jamaica Gate	-	-	-	-	250,000
Kennedy Gate	-	-	-	-	250,000
Distribution Station Over Pressure Protection	922,000	928,000	263,000	269,000	276,000
Gas System Reliability - Gas Planning /RCV Program	1,662,000	5,132,000	2,547,000	8,327,000	6,597,000
Water Intrusion	217,921	222,142	228,476	233,545	237,999
Storm Hardening - Remote Service Shutoff Valves	3,136,000	7,368,000	8,497,000	7,921,000	7,995,000
LTNY10240 - Grasmere Reliability - PM	49,664	100,000	5,142,000	-	-
LTNY11690 - LGA Backfeed - PM	-	50,000	328,000	8,654,000	-
LTNY12314 - Spring Creek - PM	4,070,467	213,467	-	-	-
LTNY10205 - MRI - PM - Main Phase 1-4	113,940,732	10,425,601	-	-	-
LTNY10205 - MRI - PM - Main Phase 5	-	54,574,399	5,729,685		
LTNY12058 - Elmhurst Reliability - PM	-	-	-	1,000,000	35,000,000
LTNY13231 - Marine Park Regulator Station - PM	99,327	-	-	-	-
LTNY11165 - Northern Queens Gas T&D - PM	513,312	13,312	-	-	-
LTNYXXXXX - Northern Line - PM	-	-	-	-	500,000
LTNYXXXXX - Northern Queens Extension - PM	-	-	-	-	100,000
Clove Lakes Transmission Line Loop (Engineering and Design	-	500,000	1,500,000	-	-
Citizens Tunnel - Upgrade	1,071,545	21,545	-	-	-
Newtown Creek	14,010,000	869,403	-	-	-
CNG - KEDNY Blanket	497,806	497,806	500,000	500,000	500,000
CNG - KEDNY Contract Closeout	-	400,000	-	-	-
CNG - NY Brooklyn (Canarsie) - Compressor Upgrade, New Controls	-	50,000	2,200,000	500,000	-
CNG - NY Brooklyn (Greenpoint) - Fueling Island Access	-	1,200,000	946,000	-	-
CNG - NY Brooklyn (Greenpoint) - New Compressors, Panels, and Controls	1,088,000	996,643	-	-	-
CNG - SI CNG Fill Station (allocate 50% to KEDLI as rent)	200,000	18,800,000	11,000,000	-	-
LNG - Blanket	2,599,086	2,648,113	2,653,763	2,712,646	2,764,373
LNG - Vaporizers 7 & 8 Replacement	100,000	500,000	10,200,000	10,127,000	3,000,000
LNG - Bulkhead Upgrade	-	700,000	-	700,000	-
LNG - Controls System Upgrade	19,865	769,865	978,000	1,712,000	-
LNG - Vaporizers 3 & 4 Replacement	21,183,000	2,000,000	-	-	-
LNG - Relocate Maintenance Area & New Control Building	-	1,406,000	6,000,000	3,000,000	1,250,000
LNG - Truck Load/Unload Station	700,000	4,813,000	12,635,000	1,032,000	-
LNG - Salt Water Pump House Upgrade	996,000	9,634,000	36,482,000	162,000	-
LNG - Geoweb Dike Replacement	1,800,000	-	-	-	-
LNG - Tank 2 Upgrade	-	-	3,705,000	5,558,000	9,623,000
LNG - Solar Panels	-	-	100,000	1,000,000	-
LNG - Liquefaction Critical Spares	49,664	949,664	-	-	-
LNG - Sub M-Sub L Interconnect	-	-	100,000	1,000,000	-
LNG - Instrument Air System Replacement	-	-	-	100,000	3,000,000
LNG - Stormwater Drainage	-	-	-	10,000	3,000,000
LNG - Hydrant & Deluge Piping Upgrade	1,800,000	4,700,000	1,500,000	-	-
LNG - Tank 1 Upgrade	-	-	-	-	-
LNG - Hi Ex Foam System	49,664	892,664	2,349,000	500,000	-
LNG - Security System Upgrades	-	-	-	100,000	2,000,000
LNG - Nitrogen System Refurbishment	-	-	-	10,000	5,000,000
LNG - Tail Gas Compressor Upgrade	669,000	100,000	5,331,000	-	-
LNG - RNG Blanket	-	200,000	200,000	200,000	200,000
LNG - Piping Insulation Replacement & Inspection	-	499,664	500,000	500,000	500,000
LNG - Boiloff Heaters/Steam Boiler Upgrade	9,933	499,933	3,000,000	3,000,000	-
LNG - Plant Outlet Drip Leg	-	10,000	500,000	-	-
LNG - Portable LNG NYC (Permitting, Site Work)	500,000	4,500,000	7,500,000	15,000,000	-
LNG - Greenpoint Vaporization Expansion (13 and 14)	1,000,000	13,000,000	21,000,000	20,000,000	-
Renewable Natural Gas (RNG) Interconnections	-	900,000	900,000	900,000	900,000
Total Reliability	194,062,012	169,171,192	178,413,285	116,249,671	104,502,598
Non-Infrastructure					
Telecomm - Radio Capital Expenditures	44,226	45,176	49,420	50,410	51,420
Telecomm - Comm site upgrades	44,089	45,039	49,420	50,410	51,420
Telecomm - Damaged Failure	11,602	11,852	13,010	13,270	13,530
Tools & Equipment - All	3,566,122	3,639,064	3,948,226	4,035,831	4,112,789
Learning and Development - Materials, Tools and Equipment	-	375,000	250,000	187,500	187,500
AMR Installation	2,934,873	2,334,873	-	-	-
Meter Testing Equipment	103,441	105,441	106,000	108,000	110,000
Automatic Meter Reading (AMR) - Replacement	3,315,998	3,385,738	3,628,460	3,701,250	3,775,270
Total Non-Infratructure	10,020,351	9,942,183	8,044,536	8,146,671	8,301,929
Total Capital Including Cost of Removal	892,247,483	885,023,011	1,025,126,413	1,032,575,164	1,052,065,132
Cost of Removal	89,135,524	88,413,799	102,410,129	103,154,259	105,101,307
Total Capital (Net of Removal)	803,111,959	796,609,212	922,716,284	929,420,905	946,963,825

Keyspan Gas East Corporation d/b/a National Grid
Direct Capital Expenditures (CAPEX and COR)

	FY20	FY21	FY22	FY23	FY24
Customer Connections					
Customer Connections - Install Main	23,597,400	21,494,500	18,535,950	14,940,144	15,238,947
Customer Connections - Install Services	26,073,660	26,454,725	26,731,610	27,266,250	27,811,570
Customer Connections - Clean Choice Program - Mair	5,920,000	-	-	-	-
Customer Connections - Clean Choice Program - Services	3,101,710	-	-	-	-
Customer Connections - Customer Contributions	(4,219,000)	(4,300,000)	(2,500,000)	(2,500,000)	(2,500,000)
Customer Connections - Meter Purchases	1,398,726	1,323,081	1,480,297	1,527,619	1,575,571
Customer Connections - Install Meter/Regulator	826,059	786,681	1,000,757	1,046,685	1,094,601
Customer Connections - Automatic Meter Reading (AMR)	933,289	882,834	927,720	957,400	987,470
Gas System Reinforcement	24,989,500	21,439,000	20,344,000	31,498,000	17,225,000
LTLI10860 Riverhead Transmission Main - PM	-	-	195,000	1,000,000	23,700,000
LTLI10985- Southeast Suffolk Infrastructure - Phase 1	600,000	25,000,000	16,600,000	-	-
LTLI10985- Southeast Suffolk Infrastructure - Phase 2	-	-	-	100,000	300,000
Total Customer Connections	83,221,344	93,080,821	83,315,334	75,836,098	85,433,159
Mandated					
CSC/Public Works - Non Reimbursable	5,246,398	5,360,398	5,536,000	5,647,000	5,686,000
CSC/Public Works - Reimbursable	5,401,132	5,517,132	5,583,000	5,694,000	5,735,000
CSC/Public Works - Reimbursements	(1,081,000)	(1,102,000)	(1,124,000)	(869,400)	(886,790)
Main Replacements (Proactive) - Leak Prone Pipe	220,251,003	233,262,396	239,093,956	245,071,304	251,198,087
Cross Bore Remediation	1,301,779	101,779	103,814	105,891	108,009
Latent Damage	504,842	514,842	530,000	540,000	550,800
Large Diameter Main Rehabilitatior	6,365,669	6,505,000	6,592,000	6,724,000	6,858,000
Main Replacements (Reactive) - Maintenance	2,240,277	2,609,202	2,710,606	2,771,361	2,826,705
Service Replacement (Reactive) - Leaks	1,854,298	1,892,745	2,081,084	2,127,260	2,167,824
Service Replacement (Reactive) - Non-Leaks - Othe	4,610,230	4,705,606	5,162,539	5,277,087	5,377,715
Restrictions for Elevated Gas Infrastructure	476,000	485,000	495,000	505,000	515,000
Buried Vent Lines	313,000	319,000	325,000	332,000	338,000
Plastic Fusion QA/QC Re-Digs	955,000	974,100	993,582	1,013,454	1,033,723
Plastic Fusion - In Process Inspections	598,500	610,470	622,679	635,132	647,834
Low Pressure Main Valve Installatior	-	50,000	51,000	52,000	53,000
Contrator Safety Inspection	-	3,613,536	11,018,676	18,470,783	18,756,036
Operator Qualification Program		652,822	461,820	470,695	480,499
Atmospheric Corrosion Inside Inspections	-	650,000	100,000	102,000	104,000
Corrosion	983,624	972,495	991,945	1,043,830	1,032,020
Pipeline Integrity - IMP	6,736,344	7,400,365	7,350,000	7,000,000	7,000,000
Pipeline Integrity - IVP	250,000	250,000	-	4,000,000	4,000,000
Pipeline Integrity -IVP - GM 9 Stewart Ave to	-	-	2,520,000	2,000,000	25,000,000
Pipeline Integrity - IVP Reactive Main Replacement	-	500,000	510,000	520,000	530,604
Damage Prevention Mark Outs and Locate	-	5,024,229	4,954,621	4,883,552	4,810,990
Valve Installations/Replacements	109,000	111,000	113,000	116,000	118,000
Meter Pitts	1,100,064	1,121,344	1,107,390	1,129,530	1,152,130
Meter Changes	2,783,685	2,861,185	3,170,000	3,249,000	3,331,000
Purchase Meters (Replacements)	3,344,407	3,411,987	3,515,400	3,585,710	3,657,420
Transmission Station Integrity	180,000	3,000,000	15,000,000	19,380,000	19,768,000
Complex Capital Delivery Initiative - Savings	-	(914,000)	(1,167,000)	(994,000)	(988,000)
Total Mandated	264,524,251	290,460,633	318,402,113	340,583,190	370,961,606
Reliability					
Gas System Reliability - Gas Planning/RCV Program	1,314,350	2,339,350	1,000,000	1,000,000	1,700,000
LTLI10652- Lynbrook- RCV QL-04	75,000	1,750,000	-	-	-
LTLI11985- Farmingdale- RCV 032583255 - PM	25,000	75,000	1,650,000	-	-
LTLI11032-Westbury- RCV 023123400 - PM	25,000	50,000	1,650,000	-	-
LTLI11715- Westbury- RCV 023123413 - PM	25,000	50,000	1,500,000	-	-
LTLI12020- Deer Park- RCV 040632167-PM	-	25,000	50,000	1,650,000	-
LTLI12021- Deer Park- RCV 040632133-PM	-	25,000	50,000	1,500,000	-
LTLI12022- Pinelawn- RCV 041025722-PM	-	25,000	50,000	1,650,000	-
LTLI10676 Elmont- RCV 007646335	-	-	25,000	50,000	1,685,000
LTLI12023- Engineering costs 2025 projects	-	-	-	-	150,000
Northwest Nassau Transmission Main & Control Valve - Phase 1	4,504,000	-	-	-	-
Northwest Nassau Transmission Main & Control Valve - Phase 2	30,705,000	79,239,000	38,000,000	2,500,000	-
Northwest Nassau Transmission Main & Control Valve - Phase 3	1,500,000	25,000,000	70,000,000	80,000,000	49,000,000
Storm Hardening - Install Remote Service Shutoff Valves	7,199,000	15,579,000	17,679,000	15,582,000	15,732,000
Water Intrusion	206,441	210,404	214,507	219,266	223,447
Gas System Control	154,530	157,430	160,130	163,750	167,770
Gas System Control - Telemetry Upgrade 3G to 4G	200,000	-	-	-	-
I&R - Reactive	265,834	270,652	260,799	266,585	271,669
I&R - Training and Test Lab	-	800,000	400,000	-	-
Heater Installation Program	-	1,504,957	1,600,000	1,600,000	1,600,000
Pressure Regulating Facilities	2,836,312	8,690,855	8,850,440	3,750,000	9,300,000
South Commack Take Station Overhaul	1,421,363	400,000	-	-	-
Rockville Centre Take Station Overhaul	100,000	4,500,000	500,000	-	-
Bay Shore Take Station Overhaul	-	400,000	2,750,000	2,500,000	-

Long Beach Gate Station Overhaul	1,200,000	-	-	-	-
ND 45	-	-	100,000	2,750,000	-
ND 02	130	-	-	-	150,000
ND 16	-	-	100,000	2,750,000	-
Riverhead Take Station	-	-	200,000	2,700,000	750,000
SL 54	2,000,000	350	-	-	-
Stewart Ave	-	-	-	200,000	3,500,000
SL 74 SL 75 Holtsville	-	-	-	-	250,000
Distribution Station Over Pressure Protection	496,000	1,746,000	1,500,000	1,500,000	276,000
Northport M&R Station Refurbishment	-	-			
System Automation	868,980	1,142,980	1,181,340	1,204,780	1,228,880
CNG - NY Hewlett - New Compressor, Controls, Storage	1,032,000	-	-	-	-
CNG - NY Brentwood - New Compressor, Controls, Storage, Dispensing	490,096	3,190,096	-	-	-
CNG - NY Riverhead - Retirement	-	500,000	500,000	-	-
CNG - NY Hicksville - Retirement	-	500,000	500,000	-	-
CNG - KEDLI Contract Closeout	-	400,000	-	-	-
CNG - KEDLI Blanket	-	500,000	500,000	500,000	500,000
CNG - Glenwood CNG Injection Facility Upgrade	500,000	6,500,000	-	-	-
CNG - Portable CNG 3rd Site	975,000	9,900,000	-	-	-
CNG - Portable CNG 4th Site	975,000	9,900,000	-	-	-
LNG - Blanket	1,054,792	1,075,085	1,098,436	1,122,808	1,144,219
LNG - Controls System Upgrade	1,527,000	6,594,000	-	-	-
LNG - AESD System	1,434,000	2,000,000	-	-	-
LNG - Storage Building	2,850,000	-	-	-	-
LNG - Security System Upgrade	-	-	-	10,000	891,000
LNG - Solar Panel Farm	-	-	-	-	150,000
LNG - Mol Sieve Refurbishment	-	-	10,000	600,000	-
LNG - Liquefaction Critical Spares	990,000	-	-	-	-
LNG - Odorant System Replacement	-	-	-	20,000	1,000,000
LNG - ReGen Heater Replacement	-	-	-	-	20,000
LNG - Boiloff Compressor System	500,000	75,000	1,000,000	1,000,000	15,292,000
LNG - SST1 & SST2 Replacement	-	-	-	-	10,000
LNG - Cyber Security Enhancements	100,000	500,000	-	-	-
LNG - Tank Upgrade	1,000,000	4,558,000	5,558,000	7,781,000	3,335,000
LNG - Analyzer Replacement 1	10,000	200,000	-	-	-
LNG - Power Center Upgrade	-	-	100,000	2,000,000	6,000,000
LNG - 4KV Cable Replacement	-	-	-	-	20,000
LNG - Nitrogen System Refurbishment	-	-	-	-	10,000
LNG - Emergency Generator Upgrade	-	-	50,000	300,000	5,000,000
LNG - Hi Ex Foam System	50,000	893,000	2,349,000	500,000	-
LNG - Liquefaction System Refurbishmen	-	-	-	50,000	3,000,000
Renewable Natural Gas (RNG) Interconnections	-	450,000	450,000	450,000	450,000
Total Reliability	68,609,828	191,716,159	161,586,651	137,870,190	122,806,985
Non-Infrastructure					
Telecomm - Comm site upgrades	47,500	48,450	49,420	50,410	51,420
Telecomm - Damaged Failure	12,500	12,750	13,010	13,270	13,530
Telecomm - Radio Capital Expenditures	48,871	49,841	50,410	51,420	52,440
Tools & Equipment - All	2,422,669	2,468,455	2,478,314	2,533,304	2,581,610
Meter Testing Equipment	198,741	208,931	216,030	227,750	240,350
Learning and Development - Materials, Tools and Equipment	-	375,000	250,000	187,500	187,500
Automatic Meter Reading (AMR) - Replacement	1,370,204	1,397,204	1,402,000	1,431,000	1,459,000
Total Non-Infratructure	4,100,485	4,560,631	4,459,184	4,494,654	4,585,850
Total Capital Including Cost of Removal	420,455,909	579,818,244	567,763,282	558,784,131	583,787,600
Cost of Removal	28,591,002	39,427,641	38,607,903	37,997,321	39,697,557
Total Capital (Net of Removal)	391,864,908	540,390,604	529,155,379	520,786,810	544,090,043

Exhibit ___ (GIOP-2SS)

KEDNY and KEDLI's Revised O&M Plans Reflecting New Customer Connections

The Brooklyn Union Gas Company d/b/a National Grid NY
Gas Safety and Gas Infrastructure and Operations Panels
Incremental Operating Expenses
Second Supplement Filing

	Rate Year 2021			Data Year 2022		
Panel / Program / Position	**Labor**	**Non-labor**	**Total**	**Labor**	**Non-labor**	**Total**
Gas Safety						
Plastic Fusion Inspection	0	102,510	102,510	0	104,560	104,560
Inside Service Line Inspection	355,447	5,973,180	6,328,627	362,560	6,150,180	6,512,740
Contractor Safety Inspection	**345,079**	**227,500**	**563,625**	**1,266,874**	**659,700**	**1,896,515**
Enhanced Inactive Accounts	3,338,549	266,450	3,604,999	6,624,004	17,082,837	23,706,841
I&R Improvements						
I&R - O&M regulator station training simulators	0	75,000	75,000	0	75,000	75,000
I&R - Site Specific Procedures	318,655	0	318,655	325,032	0	325,032
I&R - Survey & GPS map regulator station control lines	0	1,496,000	1,496,000	0	0	0
I&R - Station As-built Drawing Review	0	1,123,000	1,123,000	0	1,123,000	1,123,000
I&R Improvements Total	318,655	2,694,000	3,012,655	325,032	1,198,000	1,523,032
Enhanced Pipeline Safety Mgmt and Damage Prevention						
Damage Prevention- Damage Prevention Advisor Program	0	1,404,000	1,404,000	0	1,542,240	1,542,240
Damage Prevention FTE's - Supervisor	141,590	0	141,590	157,357	0	157,357
Damage Prevention- Markout Increases	**0**	**7,809,204**	**7,809,204**	**0**	**8,008,279**	**8,008,279**
Field Operaitons - Markout Turn Backs	0	0	0	0	0	0
Pipeline Safety Management (API 1173 Implementation)	604,351	0	604,351	616,445	0	616,445
Enhanced Pipeline Safety Mgmt and Damage Prevention Total	**745,940**	**9,213,204**	**9,959,145**	**773,802**	**9,550,519**	**10,324,321**
Materials Testing Lab	35,243	0	35,243	35,916	0	35,916
Single Meter Inspection	170,607	780,000	950,607	173,865	780,000	953,865
Gas Control SOP Training	276,618	0	276,618	282,154	0	282,154
Training (1st Responder & Field)						
Training - First Responder	235,973	50,000	285,973	240,695	50,000	290,695
Training - Field Evaluator	235,973	50,000	285,973	240,695	50,000	290,695
Training (1st Responder & Field) Total	471,945	100,000	571,945	481,390	100,000	581,390
Expanded Residential Methane Detection						
Residential Methane Detectors	0	1,674,000	1,674,000	0	1,674,000	1,674,000
Residential Methane Detectors- Education / Outreach	0	221,998	221,998	0	221,998	221,998
Expanded Residential Methane Detection Total	0	1,895,998	1,895,998	0	1,895,998	1,895,998
Enhanced High Emitter Methane Detection	43,201	250,000	293,201	44,066	255,000	299,066
Operator Qualification Program	611,616	561,597	1,173,212	623,855	675,637	1,299,492
Gas Safety Total	**6,712,899**	**22,064,439**	**28,768,385**	**10,993,518**	**38,452,430**	**49,415,889**
GIOP						
OpEx Support for Capital	314,084	0	314,084	508,107	0	508,107
D&R's related to Capital	0	1,252,867	1,252,867	0	3,114,729	3,114,729
IMP/IVP OpEx						
IVP Program PHMSA Compliance	0	0	0	0	0	0
Pipeline Integrity Support (IMP/IVP)	188,981	0	188,981	192,589	0	192,589
Pipeline Integrity- IMP (PHMSA Rules)	0	4,285,238	4,285,238	0	(853,906)	(853,906)
Pipeline Integrity- IVP (PHMSA Rules)	0	2,893,126	2,893,126	0	2,950,424	2,950,424
Capital IMP/IVP Projects Engineer (PHMSA Rules)	9,643	0	9,643	16,934	0	16,934
IMP -- ILI / ECDA (PHMSA Rules)	73,686	0	73,686	112,896	0	112,896
IMP Program Risk Model (PHMSA Rules)	32,087	0	32,087	56,349	0	56,349
IMP/IVP OpEx Total	304,396	7,178,364	7,482,760	378,768	2,096,518	2,475,286
Station Integrity						
Support PHMSA Rulemaking	142,012	0	142,012	144,854	0	144,854
Pressure Reg Engineering- Trans Station Integrity Testing	0	146,000	146,000	0	796,000	796,000
Station Integrity Total	142,012	146,000	288,012	144,854	796,000	940,854
Storm Hardening						
Storm Hardening Program	0	520,629	520,629	0	708,697	708,697
Investigate alarms, Maintain valve components	29,733	0	29,733	30,301	0	30,301
System Monitoring, valve loccation, investigate alarms	24,973	0	24,973	33,933	0	33,933
Storm Hardening Total	54,706	520,629	575,335	64,234	708,697	772,931
Fixed Factor Inspection	393,948	0	393,948	401,470	0	401,470
Research and Development	140,067	1,768,159	1,908,226	142,870	1,460,169	1,603,039
LNG - Portable Site	**0**	**0**	**0**	**0**	**0**	**0**
CNG - SI CNG Station	**0**	**0**	**0**	**0**	**17,550**	**17,550**
GIOP Total	**1,349,213**	**10,866,019**	**12,215,232**	**1,640,303**	**8,193,663**	**9,833,967**

The Brooklyn Union Gas Company d/b/a National Grid NY
Gas Safety and Gas Infrastructure and Operations Panels
Incremental Operating Expenses
Second Supplement Filing

	Data Year 2023			Data Year 2024		
Panel / Program / Position	**Labor**	**Non-labor**	**Total**	**Labor**	**Non-labor**	**Total**
Gas Safety						
Plastic Fusion Inspection	0	106,651	106,651	0	108,784	108,784
Inside Service Line Inspection	369,583	6,377,180	6,746,763	377,330	6,543,180	6,920,511
Contractor Safety Inspection	**2,150,728**	**1,074,631**	**3,175,059**	**2,194,133**	**1,025,394**	**3,169,350**
Enhanced Inactive Accounts	6,746,856	17,082,837	23,829,693	6,882,654	17,082,837	23,965,491
I&R Improvements						
I&R - O&M regulator station training simulators	0	85,000	85,000	0	85,000	85,000
I&R - Site Specific Procedures	331,328	0	331,328	338,274	0	338,274
I&R - Survey & GPS map regulator station control lines	0	0	0	0	0	0
I&R - Station As-built Drawing Review	0	1,123,000	1,123,000	0	0	0
I&R Improvements Total	331,328	1,208,000	1,539,328	338,274	85,000	423,274
Enhanced Pipeline Safety Mgmt and Damage Prevention						
Damage Prevention- Damage Prevention Advisor Program	0	1,797,888	1,797,888	0	1,948,404	1,948,404
Damage Prevention FTE's - Supervisor	160,405	0	160,405	163,767	0	163,767
Damage Prevention- Markout Increases	**0**	**8,217,956**	**8,217,956**	**0**	**8,441,440**	**8,441,440**
Field Operaitons - Markout Turn Backs	0	0	0	0	0	0
Pipeline Safety Management (API 1173 Implementation)	628,386	0	628,386	641,559	0	641,559
Enhanced Pipeline Safety Mgmt and Damage Prevention Total	**788,791**	**10,015,844**	**10,804,635**	**805,326**	**10,389,844**	**11,195,170**
Materials Testing Lab	36,579	0	36,579	37,312	0	37,312
Single Meter Inspection	177,074	780,000	957,074	180,623	780,000	960,623
Gas Control SOP Training	287,619	0	287,619	293,649	0	293,649
Training (1st Responder & Field)						
Training - First Responder	245,358	50,000	295,358	250,501	50,000	300,501
Training - Field Evaluator	245,358	50,000	295,358	250,501	50,000	300,501
Training (1st Responder & Field) Total	490,715	100,000	590,715	501,002	100,000	601,002
Expanded Residential Methane Detection						
Residential Methane Detectors	0	1,674,000	1,674,000	0	1,674,000	1,674,000
Residential Methane Detectors- Education / Outreach	0	221,998	221,998	0	221,998	221,998
Expanded Residential Methane Detection Total	0	1,895,998	1,895,998	0	1,895,998	1,895,998
Enhanced High Emitter Methane Detection	44,919	260,100	305,019	45,861	265,302	311,163
Operator Qualification Program	635,940	692,527	1,328,467	649,271	709,840	1,359,111
Gas Safety Total	**12,060,134**	**39,593,768**	**51,603,602**	**12,305,435**	**38,986,179**	**51,241,437**
GIOP						
OpEx Support for Capital	636,157	0	636,157	769,955	0	769,955
D&R's related to Capital	0	5,046,918	5,046,918	0	7,256,212	7,256,212
IMP/IVP OpEx						
IVP Program PHMSA Compliance	0	0	0	0	0	0
Pipeline Integrity Support (IMP/IVP)	196,144	0	196,144	200,075	0	200,075
Pipeline Integrity- IMP (PHMSA Rules)	0	(2,290,322)	(2,290,322)	0	2,984,646	2,984,646
Pipeline Integrity- IVP (PHMSA Rules)	0	3,009,732	3,009,732	0	3,069,967	3,069,967
Capital IMP/IVP Projects Engineer (PHMSA Rules)	17,262	0	17,262	17,624	0	17,624
IMP -- ILI / ECDA (PHMSA Rules)	115,083	0	115,083	117,495	0	117,495
IMP Program Risk Model (PHMSA Rules)	57,440	0	57,440	58,644	0	58,644
IMP/IVP OpEx Total	385,930	719,410	1,105,340	393,839	6,054,612	6,448,451
Station Integrity						
Support PHMSA Rulemaking	147,660	0	147,660	150,755	0	150,755
Pressure Reg Engineering- Trans Station Integrity Testing	0	771,000	771,000	0	787,000	787,000
Station Integrity Total	147,660	771,000	918,660	150,755	787,000	937,755
Storm Hardening						
Storm Hardening Program	0	740,655	740,655	0	903,473	903,473
Investigate alarms, Maintain valve components	30,860	0	30,860	31,479	0	31,479
System Monitoring, valve loccation, investigate alarms	34,590	0	34,590	35,315	0	35,315
Storm Hardening Total	65,450	740,655	806,105	66,794	903,473	970,267
Fixed Factor Inspection	408,881	0	408,881	417,075	0	417,075
Research and Development	145,637	1,345,602	1,491,240	148,690	1,373,819	1,522,509
LNG - Portable Site	**0**	**26,000**	**26,000**	**0**	**0**	**0**
CNG - SI CNG Station	**0**	**23,400**	**23,400**	**0**	**0**	**0**
GIOP Total	**1,789,716**	**8,672,986**	**10,462,702**	**1,947,108**	**16,375,117**	**18,322,225**

Keyspan Gas East Corporation d/b/a National Grid
Gas Safety and Gas Infrastructure and Operations Panels
Incremental Operating Expenses
Second Supplement Filing

	Rate Year 2021			Data Year 2022		
Panel / Program / Position	**Labor**	**Non-labor**	**Total**	**Labor**	**Non-labor**	**Total**
Gas Safety						
Plastic Fusion Inspection	0	203,490	203,490	0	207,560	207,560
Inside Service Line Inspection	152,129	199,814	351,943	155,121	203,864	358,985
Contractor Safety Inspection	**142,127**	**123,000**	**212,102**	**434,464**	**362,280**	**637,395**
Enhanced Inactive Accounts	218,967	44,350	263,317	334,617	412,275	746,892
I&R Improvements						
I&R - O&M regulator station training simulators	0	75,000	75,000	0	75,000	75,000
I&R - Site Specific Procedures	311,628	0	311,628	317,757	0	317,757
I&R - Survey & GPS map regulator station control lines	0	500,000	500,000	0	0	0
I&R - Station As-built Drawing Review	0	685,000	685,000	0	685,000	685,000
I&R Improvements Total	311,628	1,260,000	1,571,628	317,757	760,000	1,077,757
Enhanced Pipeline Safety Mgmt and Damage Prevention						
Damage Prevention- Damage Prevention Advisor Program	0	779,520	779,520	0	894,456	894,456
Damage Prevention FTE's - Supervisor	138,439	0	138,439	153,803	0	153,803
Damage Prevention- Markout Tickets	**0**	**2,161,952**	**2,161,952**	**0**	**2,132,445**	**2,132,445**
Field Operaitons - Markout Turn Backs	0	441,673	441,673	0	733,664	733,664
Pipeline Safety Management (API 1173 Implementation)	291,264	0	291,264	296,993	0	296,993
Enhanced Pipeline Safety Mgmt and Damage Prevention Total	**429,702**	**3,383,146**	**3,812,848**	**450,796**	**3,760,565**	**4,211,361**
Materials Testing Lab	34,053	0	34,053	34,692	0	34,692
Single Meter Inspection	2,771,164	4,000,000	6,771,164	2,823,315	4,000,000	6,823,315
Gas Control SOP Training	249,095	0	249,095	253,995	0	253,995
Training (1st Responder & Field)						
Training - First Responder	230,721	50,000	280,721	235,259	50,000	285,259
Training - Field Evaluator	230,721	50,000	280,721	235,259	50,000	285,259
Training (1st Responder & Field) Total	461,443	100,000	561,443	470,519	100,000	570,519
Expanded Residential Methane Detection						
Residential Methane Detectors	0	1,116,000	1,116,000	0	1,116,000	1,116,000
Residential Methane Detectors- Education / Outreach	0	147,999	147,999	0	147,999	147,999
Expanded Residential Methane Detection Total	0	1,263,999	1,263,999	0	1,263,999	1,263,999
Enhanced High Emitter Methane Detection	63,398	350,000	413,398	64,645	357,000	421,645
Operator Qualification Program	543,641	331,903	875,544	554,334	390,201	944,534
Gas Safety Total	**5,377,348**	**11,259,701**	**16,584,024**	**5,894,254**	**11,817,743**	**17,552,648**
GIOP						
OpEx Support for Capital	57,487	0	57,487	62,649	0	62,649
D&R's related to Capital	0	596,440	596,440	0	529,053	529,053
IMP/IVP OpEx						
IVP Program PHMSA Compliance	28,734	0	28,734	50,445	0	50,445
Pipeline Integrity Support (IMP/IVP)	272,993	0	272,993	278,117	0	278,117
Pipeline Integrity- IMP (PHMSA Rules)	0	2,133,855	2,133,855	0	678,294	678,294
Pipeline Integrity- IVP (PHMSA Rules)	0	2,219,292	2,219,292	0	2,267,983	2,267,983
Capital IMP/IVP Projects Engineer (PHMSA Rules)	8,620	0	8,620	15,134	0	15,134
IMP -- ILI / ECDA (PHMSA Rules)	65,872	0	65,872	100,890	0	100,890
IMP Program Risk Model (PHMSA Rules)	28,734	0	28,734	50,445	0	50,445
IMP/IVP OpEx Total	404,954	4,353,148	4,758,102	495,031	2,946,277	3,441,308
Station Integrity						
Support PHMSA Rulemaking	138,852	0	138,852	141,583	0	141,583
Pressure Reg Engineering- Trans Station Integrity Testing	0	517,000	517,000	0	505,000	505,000
Station Integrity Total	138,852	517,000	655,852	141,583	505,000	646,583
Storm Hardening						
Storm Hardening Program	0	876,542	876,542	0	1,308,684	1,308,684
Investigate alarms, Maintain valve components	76,639	0	76,639	78,077	0	78,077
System Monitoring, valve loccation, investigate alarms	56,000	0	56,000	76,066	0	76,066
Storm Hardening Total	132,639	876,542	1,009,181	154,144	1,308,684	1,462,828
Fixed Factor Inspection	117,958	0	117,958	120,172	0	120,172
Research and Development	136,950	956,081	1,093,031	139,643	801,846	941,489
CNG - Glenwood Site	**0**	**26,000**	**26,000**	**0**	**32,240**	**32,240**
CNG - Portable Site	**0**	**52,000**	**52,000**	**0**	**64,480**	**64,480**
CNG - SI CNG Station	**0**	**0**	**0**	**0**	**72,949**	**72,949**
GIOP Total	**988,839**	**7,377,211**	**8,366,050**	**1,113,222**	**6,260,528**	**7,373,750**

Keyspan Gas East Corporation d/b/a National Grid
Gas Safety and Gas Infrastructure and Operations Panels
Incremental Operating Expenses
Second Supplement Filing

	Data Year 2023			Data Year 2024		
Panel / Program / Position	**Labor**	**Non-labor**	**Total**	**Labor**	**Non-labor**	**Total**
Gas Safety						
Plastic Fusion Inspection	0	211,711	211,711	0	215,945	215,945
Inside Service Line Inspection	158,069	208,036	366,105	161,332	212,333	373,665
Contractor Safety Inspection	**722,437**	**589,380**	**1,068,245**	**736,818**	**574,691**	**1,176,325**
Enhanced Inactive Accounts	340,678	412,275	752,953	347,403	412,275	759,678
I&R Improvements						
I&R - O&M regulator station training simulators	0	85,000	85,000	0	85,000	85,000
I&R - Site Specific Procedures	323,796	0	323,796	330,479	0	330,479
I&R - Survey & GPS map regulator station control lines	0	0	0	0	0	0
I&R - Station As-built Drawing Review	0	685,000	685,000	0	0	0
I&R Improvements Total	323,796	770,000	1,093,796	330,479	85,000	415,479
Enhanced Pipeline Safety Mgmt and Damage Prevention						
Damage Prevention- Damage Prevention Advisor Program	0	1,115,136	1,115,136	0	1,240,848	1,240,848
Damage Prevention FTE's - Supervisor	156,726	0	156,726	159,961	0	159,961
Damage Prevention- Markout Tickets	**0**	**2,103,041**	**2,103,041**	**0**	**2,074,361**	**2,074,361**
Field Operaitons - Markout Turn Backs	0	1,083,175	1,083,175	0	1,500,620	1,500,620
Pipeline Safety Management (API 1173 Implementation)	302,636	0	302,636	308,883	0	308,883
Enhanced Pipeline Safety Mgmt and Damage Prevention Total	**459,362**	**4,301,352**	**4,760,714**	**468,844**	**4,815,829**	**5,284,673**
Materials Testing Lab	35,321	0	35,321	36,018	0	36,018
Single Meter Inspection	2,874,587	4,000,000	6,874,587	2,931,461	4,000,000	6,931,461
Gas Control SOP Training	258,821	0	258,821	264,164	0	264,164
Training (1st Responder & Field)						
Training - First Responder	239,730	50,000	289,730	244,679	50,000	294,679
Training - Field Evaluator	239,730	50,000	289,730	244,679	50,000	294,679
Training (1st Responder & Field) Total	479,460	100,000	579,460	489,357	100,000	589,357
Expanded Residential Methane Detection						
Residential Methane Detectors	0	1,116,000	1,116,000	0	1,116,000	1,116,000
Residential Methane Detectors- Education / Outreach	0	147,999	147,999	0	147,999	147,999
Expanded Residential Methane Detection Total	0	1,263,999	1,263,999	0	1,263,999	1,263,999
Enhanced High Emitter Methane Detection	65,874	364,140	430,014	67,234	371,423	438,657
Operator Qualification Program	564,868	399,956	964,824	576,528	409,955	986,482
Gas Safety Total	**6,283,274**	**12,620,848**	**18,660,549**	**6,409,637**	**12,461,449**	**18,735,902**
GIOP						
OpEx Support for Capital	63,805	0	63,805	65,087	0	65,087
D&R's related to Capital	0	574,039	574,039	0	673,919	673,919
IMP/IVP OpEx						
IVP Program PHMSA Compliance	51,404	0	51,404	52,465	0	52,465
Pipeline Integrity Support (IMP/IVP)	283,155	0	283,155	288,744	0	288,744
Pipeline Integrity- IMP (PHMSA Rules)	0	1,414,802	1,414,802	0	1,385,838	1,385,838
Pipeline Integrity- IVP (PHMSA Rules)	0	2,313,260	2,313,260	0	2,333,037	2,333,037
Capital IMP/IVP Projects Engineer (PHMSA Rules)	15,421	0	15,421	15,739	0	15,739
IMP -- ILI / ECDA (PHMSA Rules)	102,807	0	102,807	104,930	0	104,930
IMP Program Risk Model (PHMSA Rules)	51,404	0	51,404	52,465	0	52,465
IMP/IVP OpEx Total	504,191	3,728,061	4,232,253	514,343	3,718,874	4,233,217
Station Integrity						
Support PHMSA Rulemaking	144,273	0	144,273	147,251	0	147,251
Pressure Reg Engineering- Trans Station Integrity Testing	0	1,057,000	1,057,000	0	1,078,000	1,078,000
Station Integrity Total	144,273	1,057,000	1,201,273	147,251	1,078,000	1,225,251
Storm Hardening						
Storm Hardening Program	0	1,381,934	1,381,934	0	1,821,948	1,821,948
Investigate alarms, Maintain valve components	79,492	0	79,492	81,061	0	81,061
System Monitoring, valve loccation, investigate alarms	77,512	0	77,512	79,112	0	79,112
Storm Hardening Total	157,004	1,381,934	1,538,938	160,173	1,821,948	1,982,121
Fixed Factor Inspection	122,349	0	122,349	124,764	0	124,764
Research and Development	142,297	750,953	893,251	145,234	707,629	852,863
CNG - Glenwood Site	**0**	**32,240**	**32,240**	**0**	**0**	**0**
CNG - Portable Site	**0**	**64,480**	**64,480**	**0**	**0**	**0**
CNG - SI CNG Station	**0**	**1,936,338**	**1,936,338**	**0**	**1,852,953**	**1,852,953**
GIOP Total	**1,133,920**	**9,525,045**	**10,658,965**	**1,156,852**	**9,853,324**	**11,010,176**

Exhibit ___ (GIOP-3SS)

KEDNY and KEDLI's Revised Proposed Incremental FTEs

The Brooklyn Union Gas Company d/b/a National Grid NY
Incremental FTE's
Gas Safety and Gas Infrastructure and Operations Panels
Second Supplemental Filing

Panel	Program	Position	Rate Year 2021	Data Year 2022	Data Year 2023	Data Year 2024
Gas Safety	Service Line Inspection	Analyst	0.5	-	-	-
		Field Inspector	3.0	-	-	-
	Service Line Inspection Total		**3.5**	-	-	-
	Contractor Safety Inspection	**Mechanic**	**22.0**	**44.0**	**44.0**	**-**
		Supervisor	**3.0**	**6.0**	**6.0**	**-**
	Contractor Safety Inspection Total		**25.0**	**50.0**	**50.0**	**-**
	Enhanced Inactive Accounts	Call Center Representative	2.0	12.0	-	-
		Manager	-	1.0	-	-
		Mechanic	14.0	-	-	-
		Meter Service Representative	35.0	35.0	-	-
		Supervisor	2.0	1.0	-	-
	Enhanced Inactive Accounts Total		**53.0**	**49.0**	-	-
	I&R Improvements	Analyst	0.5	-	-	-
		Engineer	1.0	-	-	-
		Field Trainer	1.0	-	-	-
	I&R Improvements Total		**2.5**	-	-	-
	Gas Pipeline Safety	Pipeline Safety Management Specialis	10.0	-	-	-
		Regulatory Specialist	0.4	-	-	-
		Sr. Supervisor	1.0	-	-	-
	Gas Pipeline Safety Total		**11.4**	-	-	-
	Materials Testing Lab	Senior T&D Lead Man	0.5	-	-	-
	Materials Testing Lab Total		**0.5**	-	-	-
	Single Meter Inspection	Meter Service Representative	3.0	-	-	-
	Single Meter Inspection Total		**3.0**	-	-	-
	Gas Control SOP Training	Engineer	0.3	-	-	-
		SOP Coordinator	2.0	-	-	-
	Gas Control SOP Training Total		**2.3**	-	-	-
	Training (1st Responder & Field)	First Responder Instructor	2.0	-	-	-
		Instructor/Field Evaluator	2.0	-	-	-
	Training (1st Responder & Field) Total		**4.0**	-	-	-
	Enhanced High Emitter Methane Detection	Engineer	0.4	-	-	-
	Enhanced High Emitter Methane Detection Total		**0.4**	-	-	-
	Operator Qualification Program	Technical Inspector	11.0	-	-	-
	OQ Program Total		**11.0**	-	-	-
Gas Safety Total			**116.6**	**99.0**	**50.0**	**-**
GIOP	OpEx Support for Capital	Analyst	1.0	-	-	-
		Contract Oversight Analysts	2.0	-	-	-
		Engineer	3.0	4.0	3.0	4.0
		Inspector	2.0	1.0	1.0	1.0
		Mechanic	2.0	2.0	-	-
		Supervisor	1.0	-	-	-
		Welder	1.0	-	-	-
	OpEx Support for Capital Total		**12.0**	**7.0**	**4.0**	**5.0**
	IMP/IVP OpEx	Engineer	2.5	-	-	-
		Helper	1.0	-	-	-
		Sr. Technician	1.0	-	-	-
	IMP/IVP OpEx Total		**4.5**	-	-	-
	Station Integrity	Integrity Management Engineer	1.0	-	-	-
		Manager - Records Management	0.5	-	-	-
		Manager - Station Integrity	0.5	-	-	-
	Station Integrity Total		**2.0**	-	-	-
	Storm Hardening	Analyst	0.3	-	-	-
		Field Technician	0.3	-	-	-
	Storm Hardening Total		**0.6**	-	-	-
	Fixed Factor Inspection	Instrument Mechanic	3.0	-	-	-
	Fixed Factor Inspection Total		**3.0**	-	-	-
	Research and Development	Data Analyst	0.5	-	-	-
		Lead Engineer	0.5	-	-	-
	Research and Development Total		**1.0**	-	-	-
GIOP Total			**23.1**	**7.0**	**4.0**	**5.0**
		Grand Total Incremental	139.7	106.0	54.0	5.0
		Grand Total Cumulative Incrementa	139.7	245.7	299.7	304.7

Keyspan Gas East Corporation d/b/a National Grid
Incremental FTE's
Gas Safety and Gas Infrastructure and Operations Panels
Second Supplemental Filing

Panel	Program	Position	Rate Year 2021	Data Year 2022	Data Year 2023	Data Year 2024
Gas Safety	Service Line Inspection	Analyst	0.5	-	-	-
		Field Inspector	1.0	-	-	-
	Service Line Inspection Total		**1.5**	**-**	**-**	**-**
	Contractor Safety Inspection	**Inspector**	**13.0**	**26.0**	**26.0**	**-**
		Supervisor	**2.0**	**4.0**	**3.0**	**-**
	Contractor Safety Inspection Total		**15.0**	**30.0**	**29.0**	**-**
	Enhanced Inactive Accounts	Technician	2.0	1.0	-	-
	Enhanced Inactive Accounts Total		**2.0**	**1.0**	**-**	**-**
	I&R Improvements	Analyst	0.5	-	-	-
		Engineer	1.0	-	-	-
		Field Trainer	1.0	-	-	-
	I&R Improvements Total		**2.5**	**-**	**-**	**-**
	Gas Pipeline Safety	Pipeline Safety Management Specialist	5.0	-	-	-
		Regulatory Specialist	0.1	-	-	-
		Sr. Supervisor	1.0	-	-	-
	Gas Pipeline Safety Total		**6.1**	**-**	**-**	**-**
	Materials Testing Lab	Senior T&D Lead Man	0.5	-	-	-
	Materials Testing Lab Total		**0.5**	**-**	**-**	**-**
	Single Meter Inspection	Supervisor	1.0	-	-	-
		Technician	24.0	-	-	-
	Single Meter Inspection Total		**25.0**	**-**	**-**	**-**
	Gas Control SOP Training	Engineer	0.2	-	-	-
		SOP Coordinator	2.0	-	-	-
	Gas Control SOP Training Total		**2.2**	**-**	**-**	**-**
	Training (1st Responder & Field)	Field Evaluator	2.0	-	-	-
		First Responder Instructor	2.0	-	-	-
	Training (1st Responder & Field) Total		**4.0**	**-**	**-**	**-**
	Enhanced High Emitter Methane Detection	Engineer	0.6	-	-	-
	Enhanced High Emitter Methane Detection Total		**0.6**	**-**	**-**	**-**
	Operator Qualification Program	Technical Inspector	10.0	-	-	-
	OQ Program Total		**10.0**	**-**	**-**	**-**
Gas Safety Total			**69.4**	**31.0**	**29.0**	**-**
GIOP	OpEx Support for Capital	Contract Oversight Analysts	2.0	-	-	-
		Welder	3.0	-	-	-
	OpEx Support for Capital Total		**5.0**	**-**	**-**	**-**
	IMP/IVP OpEx	Engineer	3.5	-	-	-
		Sr. Technician	2.0	-	-	-
	IMP/IVP OpEx Total		**5.5**	**-**	**-**	**-**
	Station Integrity	Integrity Management Engineer	1.0	-	-	-
		Manager - Records Management	0.5	-	-	-
		Manager - Station Integrity	0.5	-	-	-
	Station Integrity Total		**2.0**	**-**	**-**	**-**
	Storm Hardening	Analyst	0.7	-	-	-
		Field Technician	0.7	-	-	-
	Storm Hardening Total		**1.4**	**-**	**-**	**-**
	Fixed Factor Inspection	Tester A	1.0	-	-	-
	Fixed Factor Inspection Total		**1.0**	**-**	**-**	**-**
	Research and Development	Data Analyst	0.5	-	-	-
		Lead Engineer	0.5	-	-	-
	Research and Development Total		**1.0**	**-**	**-**	**-**
GIOP Total			**15.9**	**-**	**-**	**-**
		Grand Total Incremental	85.3	31.0	29.0	-
		Grand Total Cumulative Incremental	85.3	116.3	145.3	145.3

Exhibit ___ (GIOP-4SS)

KEDLI's Revised Proactive Main Replacement (LPP) Unit Costs for FY2020-FY2024

Leak Prone Pipe - KEDLI

L02 Cost Type	FY18 $	FY18 Units	FY18	FY19	FY20	Rate Year	Data Year 1	Data Year 2	Data Year 3
Base Labor	$ 6,035,669		$ 11	$ 12	$ 12	$ 12	$ 13	$ 13	$ 13
Overtime	$ 3,409,387		$ 6	$ 7	$ 7	$ 7	$ 7	$ 7	$ 7
Other Employee Benefit	$ 2,980,140		$ 6	$ 6	$ 6	$ 6	$ 6	$ 6	$ 7
Pension and OPEB	$ 2,163,131		$ 4	$ 4	$ 4	$ 4	$ 5	$ 5	$ 5
Employee Expenses	$ 186		$ 0	$ 0	$ 0	$ 0	$ 0	$ 0	$ 0
Capital Overheads	$ 35,066,226		$ 66	$ 68	$ 70	$ 71	$ 73	$ 75	$ 77
Consultants	$ 1,922,708		$ 4	$ 4	$ 4	$ 4	$ 4	$ 4	$ 4
Contractors	$ 87,113,163		$ 165	$ 169	$ 181	$ 186	$ 190	$ 195	$ 200
Capitalized Interest	$ 2,079,198		$ 4	$ 4	$ 4	$ 4	$ 4	$ 4	$ 5
Materials	$ 3,069,104		$ 6	$ 6	$ 6	$ 6	$ 6	$ 7	$ 7
Transportation	$ 2,501,782		$ 5	$ 5	$ 5	$ 5	$ 5	$ 5	$ 5
	$ 146,340,693	**528,792**	**$ 277**	**$ 284**	**$ 299**	**$ 306**	**$ 314**	**$ 322**	**$ 330**
				Units		**768,240**	**768,240**	**768,240**	**768,240**
				Total Cost		**$ 235,190,918**	**$ 241,070,691**	**$ 247,097,458**	**$ 253,274,895**

Contractor Calculation	
FY18 (Actual)	$ 165
FY19 (Inflation 2.5%)	$ 169
60% Asplundh	$ 101
40% Other Contrators	$ 68
60% Asplundh 10.5%	$ 112
40% Other Contrators 2.5%	$ 69
FY20	$ 181

L02 Cost Type	FY18 $	FY18 Units	FY18	FY19	FY20	Rate Year	Data Year 1	Data Year 2	Data Year 3
Base Labor	$ 6,035,669		$ 11	$ 12	$ 12	$ 12	$ 13	$ 13	$ 13
Overtime	$ 3,409,387		$ 6	$ 7	$ 7	$ 7	$ 7	$ 7	$ 7
Other Employee Benefit	$ 2,980,140		$ 6	$ 6	$ 6	$ 6	$ 6	$ 6	$ 7
Pension and OPEB	$ 2,163,131		$ 4	$ 4	$ 4	$ 4	$ 5	$ 5	$ 5
Employee Expenses	$ 186		$ 0	$ 0	$ 0	$ 0	$ 0	$ 0	$ 0
Capital Overheads	$ 35,066,226		$ 66	$ 68	$ 70	$ 71	$ 73	$ 75	$ 77
Consultants	$ 1,922,708		$ 4	$ 4	$ 4	$ 4	$ 4	$ 4	$ 4
Contractors	$ 87,113,163		$ 165	$ 169	$ 179	$ 183	$ 188	$ 192	$ 197
Capitalized Interest	$ 2,079,198		$ 4	$ 4	$ 4	$ 4	$ 4	$ 4	$ 5
Materials	$ 3,069,104		$ 6	$ 6	$ 6	$ 6	$ 6	$ 7	$ 7
Transportation	$ 2,501,782		$ 5	$ 5	$ 5	$ 5	$ 5	$ 5	$ 5
	$ 146,340,693	**528,792**	**$ 277**	**$ 284**	**$ 296**	**$ 304**	**$ 311**	**$ 319**	**$ 327**
				Units		**768,240**	**768,240**	**768,240**	**768,240**
				Total Cost		**$ 233,262,396**	**$ 239,093,956**	**$ 245,071,304**	**$ 251,198,087**
				Variance		**$ (1,928,522)**	**$ (1,976,736)**	**$ (2,026,154)**	**$ (2,076,808)**

Contractor Calculation	
FY18 (Actual)	$ 165
FY19 (Inflation 2.5%)	$ 169
60% Asplundh	$ 101
40% Other Contrators	$ 68
60% Asplundh 7.9%	$ 109
40% Other Contrators 2.5%	$ 69
FY20	$ 179

Exhibit ___ (GIOP-5SS)

Worksheets Summarizing Changes to Capital Plan

The Brooklyn Union Gas Company and Keyspan Gas East Corporation d/b/a National Grid
NY PSC Case 19-G-0309 and 19-G-0310
Capital Adjustments included in Company Second Supplemental Filing

	Capital Second Supplement Adjustments				
	FY2020	FY2021	FY2022	FY2023	FY2024
Capital Project Changes Related to Gas Connections:					
KEDNY	54,746,775	70,597,968	108,506,406	128,639,942	102,390,410
KEDLI	29,624,190	92,898,963	64,300,061	80,938,144	88,027,274
Total Capital Project Changes Related to Gas Connections	84,370,965	163,496,931	172,806,467	209,578,086	190,417,684
Capital Project Changes Related to Renegotiated Contractor Rate Changes:					
KEDLI	0	3,095,707	2,977,886	2,857,398	2,734,182
Total Capital Project Changes Related to Renegotiated Contractor Rate Changes	0	3,095,707	2,977,886	2,857,398	2,734,182
Total Capital Project Changes in Companies Second Supplemental Filing	84,370,965	166,592,637	175,784,353	212,435,484	193,151,866

Exhibit_____(GIOP-5SS)
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The Brooklyn Union Gas Company d/b/a National Grid
NY PSC Case 19-G-0309
Company Second Supplemental Filing

KEDNY Capital Project Changes Related to Gas Connections for Second Supplemental Filing

	Corrections & Updates					Rebuttal Adjustments					Rebuttal					Second Supplement Adjustments					Second Supplement				
	FY2020	FY2021	FY2022	FY2023	FY2024	FY2020	FY2021	FY2022	FY2023	FY2024	FY2020	FY2021	FY2022	FY2023	FY2024	FY2020	FY2021	FY2022	FY2023	FY2024	FY2020	FY2021	FY2022	FY2023	FY2024
Gas Customer Connection Project Changes																									
Customer Connections - Install Main	21,146,720	21,729,722	22,538,940	22,989,719	23,449,513	(14,160,720)	(14,592,998)	(18,100,635)	(18,183,242)	(18,015,054)	6,986,000	7,136,724	4,438,305	4,806,477	5,434,459	14,160,720	14,592,998	18,100,635	18,183,242	18,015,054	21,146,720	21,729,722	22,538,940	22,989,719	23,449,513
Customer Connections - Install Services	24,785,380	25,488,092	27,925,611	28,484,123	29,053,806	(10,096,105)	(14,680,772)	(18,430,856)	(18,820,048)	(18,986,577)	14,689,275	10,807,320	9,494,755	9,664,075	10,067,229	10,096,105	14,680,772	18,430,856	18,820,048	18,986,577	24,785,380	25,488,092	27,925,611	28,484,123	29,053,806
Customer Connections - Customer Contributions	(2,307,000)	(2,352,000)	(2,403,000)	(2,456,000)	(2,503,000)	0	(2,088,896)	(2,276,482)	(2,834,954)	(3,525,063)	(2,307,000)	(4,440,896)	(4,679,482)	(5,290,954)	(6,028,063)	0	2,088,896	2,276,482	2,834,954	3,525,063	(2,307,000)	(2,352,000)	(2,403,000)	(2,456,000)	(2,503,000)
Customer Connections - Meter Purchases	1,811,750	1,847,990	1,884,950	1,922,640	1,961,100	0	(1,597,536)	(1,629,486)	(1,662,067)	(1,695,316)	1,811,750	250,454	255,464	260,573	265,784	0	1,597,536	1,629,486	1,662,067	1,695,316	1,811,750	1,847,990	1,884,950	1,922,640	1,961,100
Customer Connections - Install Meter/Regulator	1,232,673	1,257,700	1,336,904	1,363,642	1,390,915	(516,750)	(909,618)	(1,032,663)	(1,076,498)	(1,104,386)	715,923	348,082	304,241	287,144	286,529	516,750	909,618	1,032,663	1,076,498	1,104,386	1,232,673	1,257,700	1,336,904	1,363,642	1,390,915
Customer Connections - Automatic Meter Reading (AMR)	1,042,000	1,062,090	1,083,330	1,105,000	1,127,100	0	(901,942)	(919,979)	(938,382)	(957,149)	1,042,000	160,148	163,351	166,618	169,951	0	901,942	919,979	938,382	957,149	1,042,000	1,062,090	1,083,330	1,105,000	1,127,100
Gas System Reinforcement	45,382,000	13,641,000	61,716,000	84,342,000	64,031,000	(4,538,200)	(6,745,500)	(45,837,000)	(46,590,750)	(48,023,250)	40,843,800	6,895,500	15,879,000	37,751,250	16,007,750	4,538,200	6,745,500	45,837,000	46,590,750	48,023,250	45,382,000	13,641,000	61,716,000	84,342,000	64,031,000
Gas System Reinforcement -LTNYXXXXX-Jamaica Inlet - PM	0	0	0	0	0	100,000	520,000	9,913,000			100,000	520,000	9,913,000	0	0	(100,000)	(520,000)	(9,913,000)	0	0	0	0	0	0	0
Total Customer Connection Project Changes	93,093,523	62,674,594	114,082,735	137,751,124	118,510,433	(29,211,775)	(40,997,262)	(78,314,101)	(90,105,941)	(92,306,795)	63,881,748	21,677,332	35,768,634	47,645,183	26,203,638	29,211,775	40,997,262	78,314,101	90,105,941	92,306,795	93,093,523	62,674,594	114,082,735	137,751,124	118,510,433
																0	0	0	0	0					
Mandated Project Changes																									
Contractor Safety Inspections	0	5,370,628	16,363,614	27,786,267	28,182,040	0	(87,706)	(267,305)	(454,001)	(460,615)	0	5,282,922	16,096,309	27,332,266	27,721,425	0	87,706	267,305	454,001	460,615	0	5,370,628	16,363,614	27,786,267	28,182,040
Total Mandated Project Changes	0	5,370,628	16,363,614	27,786,267	28,182,040	0	(87,706)	(267,305)	(454,001)	(460,615)	0	5,282,922	16,096,309	27,332,266	27,721,425	0	87,706	267,305	454,001	460,615	0	5,370,628	16,363,614	27,786,267	28,182,040
																0	0	0	0	0					
Reliability Project Changes																									
LTNY10205 - MRI - PM - Main Phase 1-4	88,940,732	35,425,601	0	0	0	0	0	0	0	0	88,940,732	35,425,601	0	0	0	25,000,000	(25,000,000)				113,940,732	10,425,601	0	0	0
LTNY10205 - MRI - PM - Main Phase 5	0	39,574,399	20,729,685	0	0	0	0	0	0	0	0	39,574,399	20,729,685	0	0		15,000,000	(15,000,000)			0	54,574,399	5,729,685	0	0
LTNY13231 - Marine Park Regulator Station - PM	99,327	999,327	22,769,000	0	0	0	(999,327)	(22,769,000)	0	0	99,327	0	0	0	0	0	0	0	0	0	99,327	0	0	0	0
LNG - Truck Load/Unload Station	1,865,000	2,100,000	12,265,000	510,000	0	0	0	0	0	0	1,865,000	2,100,000	12,265,000	510,000	0	(1,165,000)	2,713,000	370,000	522,000		700,000	4,813,000	12,635,000	1,032,000	0
LNG - Tank 2 Upgrade	0	0	100,000	1,500,000	1,500,000	0	0	0	0	(1,500,000)	0	0	100,000	1,500,000	0	0	0	3,605,000	4,058,000	9,623,000	0	0	3,705,000	5,558,000	9,623,000
LNG - Tank 1 Upgrade	0	0	50,000	1,500,000	500,000	0	0	0	0	(500,000)	0	0	50,000	1,500,000	0	0	0	(50,000)	(1,500,000)	0	0	0	0	0	0
Clove Lakes Transmission Line Loop (Engineering and Design)	0	0	0	0	0	0	0	0	0	0						0	500,000	1,500,000	0	0	0	500,000	1,500,000	0	0
LNG - Portable LNG NYC (Permitting, Site Work)	0	0	0	0	0	0	0	0	0	0						500,000	4,500,000	7,500,000	15,000,000	0	500,000	4,500,000	7,500,000	15,000,000	0
LNG - Greenpoint Vaporization Expansion (13 and 14)	0	0	0	0	0	0	0	0	0	0						1,000,000	13,000,000	21,000,000	20,000,000	0	1,000,000	13,000,000	21,000,000	20,000,000	0
CNG - SI CNG Fill Station (allocate 50% to KEDLI as rent)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	200,000	18,800,000	11,000,000	0	0	200,000	18,800,000	11,000,000	0	0
Total Reliability Project Changes	90,905,059	78,099,327	55,913,685	3,510,000	2,000,000	0	(999,327)	(22,769,000)	0	(2,000,000)	90,905,059	77,100,000	33,144,685	3,510,000	0	25,535,000	29,513,000	29,925,000	38,080,000	9,623,000	116,440,059	106,613,000	63,069,685	41,590,000	9,623,000
Total Capital Project Changes Related to Gas Connection	183,998,582	146,144,549	186,360,034	169,047,391	148,692,473	(29,211,775)	(42,084,295)	(101,350,406)	(90,559,942)	(94,767,410)	154,786,807	104,060,254	85,009,628	78,487,449	53,925,063	54,746,775	70,597,968	108,506,406	128,639,942	102,390,410	209,533,582	174,658,222	193,516,034	207,127,391	156,315,473

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Keyspan Gas East Corporation d/b/a National Grid
NY PSC Case 19-G-0310
Company Second Supplemental Filing

	Corrections & Updates					Rebuttal Adjustments					Rebuttal					Second Supplement Adjustments					Second Supplement				
	FY2020	FY2021	FY2022	FY2023	FY2024	FY2020	FY2021	FY2022	FY2023	FY2024	FY2020	FY2021	FY2022	FY2023	FY2024	FY2020	FY2021	FY2022	FY2023	FY2024	FY2020	FY2021	FY2022	FY2023	FY2024
KEDLI Capital Project Changes Related to Gas Connections:																									
Gas Customer Connection Project Changes																									
Customer Connections - Install Main	23,597,400	21,494,500	18,535,950	14,940,144	15,238,947	(11,482,400)	(19,922,507)	(17,635,099)	(14,426,507)	(14,961,307)	12,115,000	1,571,993	900,851	513,637	277,640	11,482,400	19,922,507	17,635,099	14,426,507	14,961,307	23,597,400	21,494,500	18,535,950	14,940,144	15,238,947
Customer Connections - Install Services	26,073,660	26,454,725	26,731,610	27,266,250	27,811,570	(12,892,840)	(21,255,266)	(21,323,534)	(22,391,935)	(22,853,147)	13,180,820	5,199,459	5,408,076	4,874,315	4,958,423	12,892,840	21,255,266	21,323,534	22,391,935	22,853,147	26,073,660	26,454,725	26,731,610	27,266,250	27,811,570
Customer Connections - Clean Choice Program - Main	23,125,000	20,790,000	18,314,100	15,730,848	13,036,940	(17,205,000)	(20,790,000)	(18,314,100)	(15,730,848)	(13,036,940)	5,920,000	0	0	0	0	0	0	0	0	0	5,920,000	0	0	0	0
Customer Connections - Clean Choice Program - Services	5,582,240	4,768,560	4,197,879	3,602,464	2,981,553	(2,480,530)	(4,768,560)	(4,197,879)	(3,602,464)	(2,981,553)	3,101,710	0	0	0	0	0	0	0	0	0	3,101,710	0	0	0	0
Customer Connections - Customer Contribution	(9,219,000)	(4,300,000)	(2,500,000)	(2,500,000)	(2,500,000)	5,000,000	(619,432)	(2,557,083)	(2,698,351)	(2,803,003)	(4,219,000)	(4,919,432)	(5,057,083)	(5,198,351)	(5,303,003)	0	619,432	2,557,083	2,698,351	2,803,003	(4,219,000)	(4,300,000)	(2,500,000)	(2,500,000)	(2,500,000)
Customer Connections - Meter Purchases	1,398,726	1,429,086	1,579,450	1,611,040	1,643,260	0	(1,429,086)	(1,579,450)	(1,611,040)	(1,643,260)	1,398,726	0	0	0	0	0	1,323,081	1,480,297	1,527,619	1,575,571	1,398,726	1,323,081	1,480,297	1,527,619	1,575,571
Customer Connections - Install Meter/Regulator	826,059	860,997	1,067,141	1,104,491	1,143,148	0	(722,832)	(925,233)	(1,004,407)	(1,066,482)	826,059	138,165	141,908	100,084	76,666	0	648,516	858,849	946,601	1,017,935	826,059	786,681	1,000,757	1,046,685	1,094,601
Customer Connections - Automatic Meter Reading (AMR)	933,289	953,319	989,990	1,009,790	1,029,980	0	(953,319)	(989,990)	(1,009,790)	(1,029,980)	933,289	0	0	0	0	0	882,834	927,720	957,400	987,470	933,289	882,834	927,720	957,400	987,470
Gas System Reinforcement	24,989,500	21,439,000	20,344,000	31,498,000	17,225,000	(2,498,950)	(12,661,500)	(16,850,000)	(25,855,750)	(13,293,750)	22,490,550	8,777,500	3,494,000	5,642,250	3,931,250	2,498,950	12,661,500	16,850,000	25,855,750	13,293,750	24,989,500	21,439,000	20,344,000	31,498,000	17,225,000
LTL110860 Riverhead Transmission Main - PM	0	0	195,000	1,000,000	23,700,000	0	0	(195,000)	(1,000,000)	(23,700,000)	0	0	0	0	0	0	0	195,000	1,000,000	23,700,000	0	0	195,000	1,000,000	23,700,000
LTL110985- Southeast Suffolk Infrastructure - Phase 1	600,000	20,000,000	21,600,000	0	0	0	0	0	0	0	600,000	20,000,000	21,600,000	0	0		5,000,000	(5,000,000)			600,000	25,000,000	16,600,000	0	0
LTL110985- Southeast Suffolk Infrastructure - Phase 2	0	0	0	100,000	300,000	0	0	0	(100,000)	(300,000)	0	0	0	0	0	0	0	0	100,000	300,000	0	0	0	100,000	300,000
Total Customer Connection Project Changes	97,906,874	113,890,187	111,055,120	95,363,027	101,610,398	(41,559,720)	(83,122,502)	(84,567,368)	(89,431,092)	(97,669,423)	56,347,154	30,767,685	26,487,752	5,931,935	3,940,975	26,874,190	62,313,136	56,827,582	69,904,163	81,492,184	83,221,344	93,080,821	83,315,334	75,836,098	85,433,159
Mandated Project Changes																									
Contractor Safety Inspections	0	3,613,536	11,018,676	18,470,783	18,756,036	0	(627,827)	(1,914,479)	(3,252,981)	(3,200,090)	0	2,985,709	9,104,197	15,217,802	15,555,946	0	627,827	1,914,479	3,252,981	3,200,090	0	3,613,536	11,018,676	18,470,783	18,756,036
Total Mandated Project Changes	0	3,613,536	11,018,676	18,470,783	18,756,036	0	(627,827)	(1,914,479)	(3,252,981)	(3,200,090)	0	2,985,709	9,104,197	15,217,802	15,555,946	0	627,827	1,914,479	3,252,981	3,200,090	0	3,613,536	11,018,676	18,470,783	18,756,036
Reliability Project Changes																									
LNG - Tank Upgrade	700,000	900,000	4,113,000	22,039,000	36,483,000	0	0	(4,113,000)	(22,039,000)	(36,483,000)	700,000	900,000	0	0	0	300,000	3,658,000	5,558,000	7,781,000	3,335,000	1,000,000	4,558,000	5,558,000	7,781,000	3,335,000
CNG - Glenwood CNG Injection Facility Upgrade	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	500,000	6,500,000	0	0	0	500,000	6,500,000	0	0	0
CNG - Portable CNG 3rd Site	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	975,000	9,900,000	0	0	0	975,000	9,900,000	0	0	0
CNG - Portable CNG 4th Site	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	975,000	9,900,000	0	0	0	975,000	9,900,000	0	0	0
Total Reliability Project Changes	700,000	900,000	4,113,000	22,039,000	36,483,000	0	0	(4,113,000)	(22,039,000)	(36,483,000)	700,000	900,000	0	0	0	2,750,000	29,958,000	5,558,000	7,781,000	3,335,000	3,450,000	30,858,000	5,558,000	7,781,000	3,335,000
Total Capital Project Changes Related to Gas Connections	98,606,874	118,403,723	126,186,796	135,872,810	156,849,434	(41,559,720)	(83,750,329)	(90,594,847)	(114,723,073)	(137,352,513)	57,047,154	34,653,394	35,591,949	21,149,737	19,496,921	29,624,190	92,898,963	64,300,061	80,938,144	88,027,274	86,671,344	127,552,357	99,892,010	102,087,881	107,524,195
KEDLI Capital Project Changes Related to Renegotiated Contractor Rate Changes:																									
Main Replacements (Proactive) - Leak Prone Pipe	220,251,003	235,190,918	241,070,691	247,097,458	253,274,895	0	0	0	0	0	220,251,003	235,190,918	241,070,691	247,097,458	253,274,895	0	(1,928,522)	(1,976,736)	(2,026,154)	(2,076,808)	220,251,003	233,262,396	239,093,956	245,071,304	251,198,087
Damage Prevention Mark Outs and Locate Change (Note 1)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5,024,229	4,954,621	4,883,552	4,810,990	0	5,024,229	4,954,621	4,883,552	4,810,990
Total Capital Project Changes Related to Renegotiated Contr Rate Changes	220,251,003	235,190,918	241,070,691	247,097,458	253,274,895	0	0	0	0	0	220,251,003	235,190,918	241,070,691	247,097,458	253,274,895	0	3,095,707	2,977,886	2,857,398	2,734,182	220,251,003	238,286,625	244,048,577	249,954,856	256,009,077
Total Capital Project Changes in Company Rebuttal Supplemental Filing																29,624,190	95,994,669	67,277,947	83,795,542	90,761,456					

Note 1 - In Corrections and Updates Damage Prevention Mark Outs and Locate are embedded in all of the capital programs involving excavation

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Exhibit ___ (GIOP-6SS)

Worksheets Summarizing Changes to O&M Plan

The Brooklyn Union Gas Company and Keyspan Gas East Corporation d/b/a National Grid NY
Second Supplemental Filing
Changes to Other Initiatives
Incremental O&M and FTE's

Unit: ($000)

Company/Cost Type/Program		Rebuttal Filing				Second Supplemental Filing				Variance			
		RY2021	DY2022	DY2023	DY2024	RY2021	DY2022	DY2023	DY2024	RY2021	DY2022	DY2023	DY2024
KEDNY													
O&M Expense													
Contractor Safety Inspection	Labor & OH's	$339.8	$1,247.4	$2,117.6	$2,160.4	$345.1	$1,266.9	$2,150.7	$2,194.1	$5.3	$19.5	$33.1	$33.8
	Non-labor	$223.9	$649.1	$1,057.4	$1,009.0	$227.5	$659.7	$1,074.6	$1,025.4	$3.6	$10.6	$17.2	$16.4
	Sub-total	$563.6	$1,896.5	$3,175.1	$3,169.4	$572.6	$1,926.6	$3,225.4	$3,219.5	$9.0	$30.1	$50.3	$50.2
Damage Prevention Mark Outs - Renegotiated Contract	Non-labor	$980.2	$1,232.8	$1,560.8	$1,986.7	$7,809.2	$8,008.3	$8,218.0	$8,441.4	$6,829.0	$6,775.5	$6,657.1	$6,454.7
LNG - Portable Site	Non-labor	$0.0	$0.0	$0.0	$0.0	$0.0	$0.0	$26.0	$0.0	$0.0	$0.0	$26.0	$0.0
CNG - SI CNG Station O&M	Non-labor	$0.0	$0.0	$0.0	$0.0	$0.0	$17.6	$23.4	$0.0	$0.0	$17.6	$23.4	$0.0
	Total	$1,543.9	$3,129.3	$4,735.9	$5,156.1	$8,381.8	$9,952.4	$11,492.7	$11,661.0	$6,837.9	$6,823.1	$6,756.8	$6,504.9
Incremental FTE's													
Contractor Safety Inspection													
	Inspector	21.6	43.2	43.2	0	22	44	44	0	0.4	0.8	0.8	0.0
	Supervisor	3	6	6	0	3	6	6	0	0.0	0.0	0.0	0.0
	Total	24.6	49.2	49.2	0	24.6	49.2	49.2	0	0.4	0.8	0.8	0.0
KEDLI													
O&M Expense													
Contractor Safety Inspection													
	Labor & OH's	$113.7	$347.6	$589.9	$601.6	$142.1	$434.5	$722.4	$736.8	$28.4	$86.9	$132.6	$135.2
	Non-labor	$98.4	$289.8	$478.4	$574.7	$123.0	$362.3	$589.4	$574.7	$24.6	$72.5	$111.0	$0.0
	Sub-total	$212.1	$637.4	$1,068.2	$1,176.3	$265.1	$796.7	$1,311.8	$1,311.5	$53.0	$159.3	$243.6	$135.2
Damage Prevention Mark Outs - Renegotiated Contract	Non-labor	$224.8	$324.8	$442.8	$583.8	$2,162.0	$2,132.4	$2,103.0	$2,074.4	$1,937.2	$1,807.6	$1,660.3	$1,490.6
CNG - Glenwood Site	Non-labor	$0.0	$0.0	$0.0	$0.0	$26.0	$32.2	$32.2	$0.0	$26.0	$32.2	$32.2	$0.0
CNG - Portable Site	Non-labor	$0.0	$0.0	$0.0	$0.0	$52.0	$64.5	$64.5	$0.0	$52.0	$64.5	$64.5	$0.0
CNG - SI CNG Station O&M	Non-labor	$0.0	$0.0	$0.0	$0.0	$0.0	$17.6	$23.4	$0.0	$0.0	$17.6	$23.4	$0.0
CNG - SI CNG Station (Rent)	Non-labor	$0.0	$0.0	$0.0	$0.0	$0.0	$55.4	$1,912.9	$1,853.0	$0.0	$55.4	$1,912.9	$1,853.0
	Total	$436.9	$962.2	$1,511.0	$1,760.1	$2,505.1	$3,098.9	$5,447.9	$5,238.8	$2,068.2	$2,136.6	$3,936.9	$3,478.7
Incremental FTE's													
Contractor Safety Inspection	Inspector	10.4	20.8	20.8	0	13	26	26	0	2.6	5.2	5.2	0.0
	Supervisor	1.6	3.2	3.2	0	2	4	3	0	0.4	0.8	(0.2)	0.0
	Total	12	24	24	0	12	24	24	0	3.0	6.0	5.0	0.0

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